Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/02/2007
PI Name:	Kuzmich, Alex Ph.D.		
Project Title:	Entangled atomic clock		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Atlanta	**State:**	GA
Zip Code:	30332-0100	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	7	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per PI (7/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kennedy, T.A.Brian (School of Physics)		
Grant/Contract No.:	GSN 4516		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A four-year program of experimental and theoretical research on development of entanglement-enhanced atomic clock will be undertaken. Atomic entanglement will be generated via collective QND measurements of populations of the two levels involved in the atomic clock transition. In the first phase of the research, experimental and theoretical investigations will be done with a cold sample of atoms typical of present-day atomic clocks. In the second stage, we will investigate BEC-based entangled atomic clock. For the latter case we will pursue theoretical and experimental investigations of the issues of stability of spatial modes and the impact of the collisional frequency shift on the entanglement-enhanced performance of such a clock. We intend to experimentally implement a quantum non-demolition interaction between our atomic ensemble and a light field. Measurement of the light field amounts to a quantum non-demolition measurement of the atoms, creating an entangled state. The essential new idea we propose is the use of a single off-resonant light field which when correctly tuned with respect to the hyperfine levels measures the population difference of the levels levels $|1\rangle $ and $|2\rangle $ associated with the clock transition. Moreover, under these conditions the interaction conserves the population difference of the atomic sample, a vital element in our experimental measurement strategy. Another objective of the research program is to investigate methods which allow cancellation of classical noise sources, such as those associated with distribution of coupling strengths seen by an atomic ensemble in a spatially varying field mode. If uncorrected, such noise could entirely mask entanglement, precluding the possible enhancement in accuracy of an atomic clock. We propose the use of ``matched" observables, which enable large amounts of entanglement to be extracted under these circumstances of apparently weak entanglement. We will investigate the concept of matched observables in the contexts of both cold non-degenerate and BEC-based atomic clocks. The proposed work will significantly impact possible future microgravity atomic clock experiments. Those experiments will be attempting to reach higher accuracy than ground-based atomic clocks. One avenue to even higher accuracy is to increase the number of atoms and/or interaction time. However, this avenue may have already been pushed close to its limits. The development of an entanglement-enhanced atomic clock would represent a major breakthrough and pave the way for further improvements. Measurement-induced entanglement on the atomic clock transition may very well become an integral part of the space-based atomic clock of the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research on quantum repeaters relates to a potentially secure information transfer system using quantum mechanics.
Task Progress:	We have made significant advances in the generation, distribution, and storage of qubit entanglement using laser manipulation of atomic ensembles, including atom-photon entanglement and matter-light qubit conversion, Bell inequality violation between a collective atomic qubit and a photon, and light-matter qubit conversion and entanglement of remote atomic qubits, and the foundation of quantum telecommunications. Subsequently, we have proposed and demonstrated the method for generating entangled photons at telecommunication wavelength suitable for long distance quantum repeater protocols and based on atomic cascade emission. Also in connection to such protocols we describe theoretically and demonstrate experimentally Hong-Ou-Mandel interference of single photons produced by independent sources of ultracold atomic rubidium. We also proposed and demonstrated an inteferometrically robust atomic qubit based on a mixture of two rubidium isotopes and entangled with independent frequency components of electromagnetic field. This contrasts with our earlier experiments in which quantum information was carried in polarization states of the light field. We developed a new theoretical architecture for a long-distance quantum repeater system based on entanglement connection and distibution which takes account of two most significant limiting factors in current physical implementations: the need to use telecommunication wavelength photons to communicate over long distances without significant loss, and the severe limitation on repeater memory element lifetimes. We have in mind systems where the memory elements consist of cold atomic ensembles. The current record on the memory lifetime, demonstrated by our group, is 31 microseconds. Since this limits the available communication time between memory elements, determined by the speed of light in the optical fiber communication channel, a new, multiplexed, topology of memory elements was developed to enhance the probability of entanglement connection between distinct memory elements. The basic goal is to develop a long distance quantum communication system within the constraints set by achievable memory element lifetimes. Ongoing work involves optimization of the system design taking into account the microscopic atom-photon interaction physics on which it ultimately relies. We are currently developing experimental methods to realize multiplexed memory elements by means of spatially separate elements within a single ultracold atomic ensemble. This can be achieved by using acousto-optic modulation to separately address different regions of the vapor using light signals to perform entanglement connection and its verification.
Bibliography Type:	Description: (Last Updated: 02/13/2008)
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